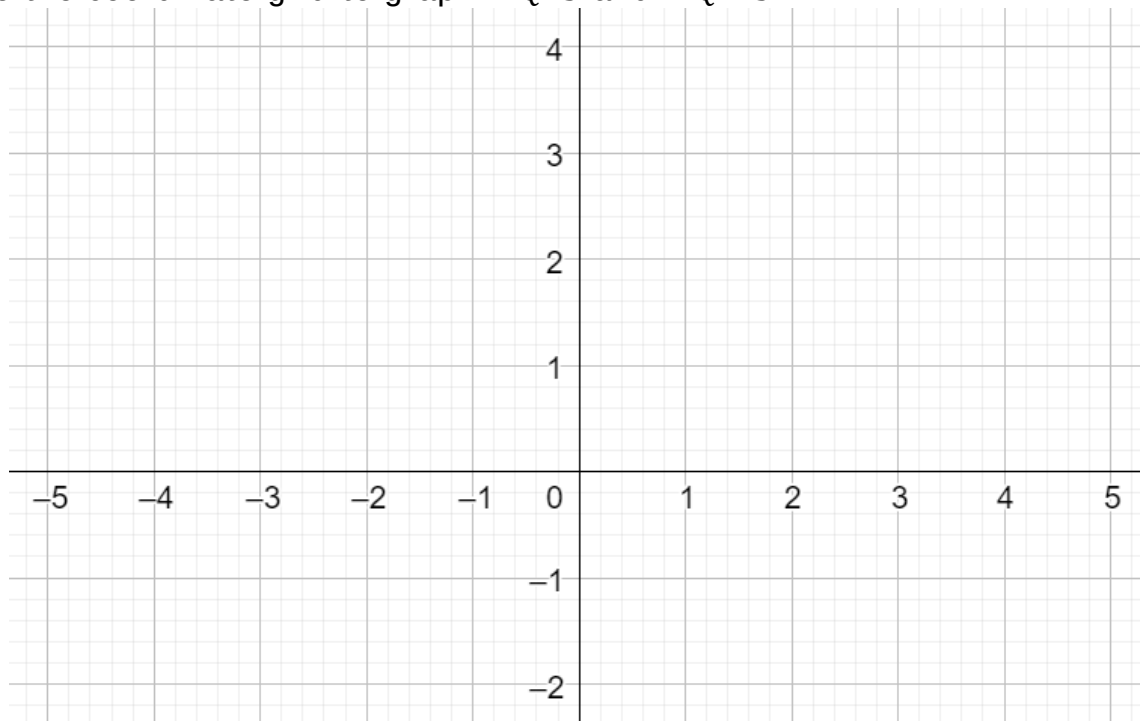


The coordinates of $\triangle QRS$ and $\triangle Q'R'S'$ are shown in the table, where $\triangle QRS$ is the preimage:

$\triangle QRS$		
Vertex	x	y
Q	-1	2
R	-3	1
S	-4	4

$\triangle Q'R'S'$		
Vertex	x	y
Q'	2	1
R'	1	3
S'	4	4

1. Use the coordinate grid to graph $\triangle QRS$ and $\triangle Q'R'S'$.



2. What type of transformation was applied to the preimage $\triangle QRS$?
3. Was distance preserved in this transformation?
4. $\triangle Q'R'S'$ is further transformed by dilating with a scale factor of $\frac{1}{2}$ centered at the origin to create $\triangle Q''R''S''$. Graph $\triangle Q''R''S''$ on the coordinate grid above.
5. Was distance preserved in this second transformation from $\triangle Q'R'S'$ to $\triangle Q''R''S''$?

Complete the following statements about the transformation of preimage XYZ :

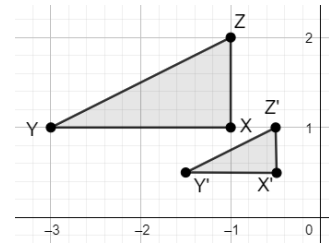
6. Image $X'Y'Z'$ was created by

- ☐ translation
- ☐ reflection
- ☐ rotation
- ☐ dilation

This transformation

- ☐ does
- ☐ does not

preserve distance.



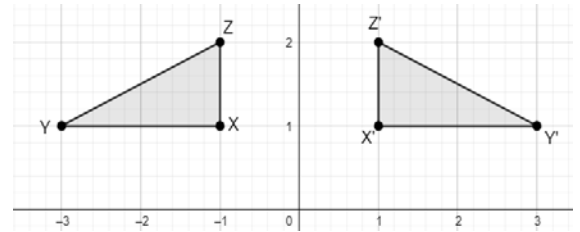
7. Image $X'Y'Z'$ was created by

- ☐ translation
- ☐ reflection
- ☐ rotation
- ☐ dilation

This transformation

- ☐ does
- ☐ does not

preserve distance.



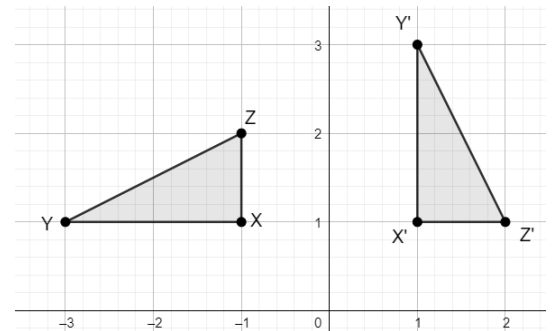
8. Image $X'Y'Z'$ was created by

- ☐ translation
- ☐ reflection
- ☐ rotation
- ☐ dilation

This transformation

- ☐ does
- ☐ does not

preserve distance.



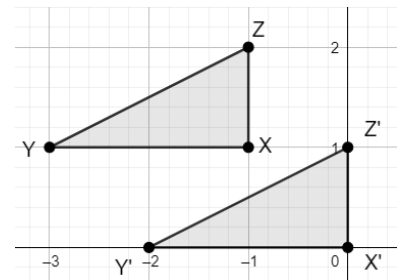
9. Image $X'Y'Z'$ was created by

- ☐ translation
- ☐ reflection
- ☐ rotation
- ☐ dilation

This transformation

- ☐ does
- ☐ does not

preserve distance.



10. Image MLH was created either by

or by

- ☐ translation
- ☐ reflection
- ☐ rotation
- ☐ dilation

- ☐ translation
- ☐ reflection
- ☐ rotation
- ☐ dilation

Both of these transformations

- ☐ do
- ☐ do not

preserve distance.

